

WEB TECHNOLOGY
UNIT IV - REPRESENTING WEB DATA
ASSESSMENT I: DATA INTERPRETATION

Duration	60 Minutes	Maximum Marks	30
Mode	Individual Assessment	Unit	IV

Instructions

- Study the given XML dataset carefully before answering the questions.
- Write valid XPath expressions for the specified data-selection requirements.
- Develop the required XSLT transformation using correct XML/XSLT syntax.
- Interpret the extracted information based on the given academic scenario.
- Appropriate assumptions may be made only when they are clearly stated.

Scenario: University Course Enrollment Analysis

A university maintains information about the courses offered during a semester in XML format. The Academic Office wants to analyse course enrollment, credit distribution, and course type before preparing the semester workload report. Use the following XML data for all questions in this assessment.

Given XML Dataset

```
<?xml version="1.0" encoding="UTF-8"?>
<courses>
  <course id="C101">
    <code>WEB301</code>
    <name>Web Technology</name>
    <faculty>Dr. Arun</faculty>
    <students>58</students>
    <credits>4</credits>
    <type>Theory</type>
  </course>
  <course id="C102">
    <code>AI302</code>
    <name>Artificial Intelligence</name>
    <faculty>Dr. Meena</faculty>
    <students>72</students>
    <credits>4</credits>
    <type>Theory</type>
  </course>
  <course id="C103">
```

```

<code>WEB303</code>
<name>Web Technology Laboratory</name>
<faculty>Dr. Ravi</faculty>
<students>36</students>
<credits>2</credits>
<type>Practical</type>
</course>
<course id="C104">
  <code>ML304</code>
  <name>Machine Learning</name>
  <faculty>Dr. Priya</faculty>
  <students>64</students>
  <credits>4</credits>
  <type>Theory</type>
</course>
<course id="C105">
  <code>DB305</code>
  <name>Database Systems</name>
  <faculty>Dr. Kumar</faculty>
  <students>42</students>
  <credits>3</credits>
  <type>Theory</type>
</course>
</courses>

```

Question 1: Interpret the XML Structure - 5 Marks

a. Identify the root element.

b. Identify the repeating record element.

c. Identify the attribute used to uniquely identify each course.

d. Identify the elements that represent numeric information.

e. State whether the XML document is structurally well-formed and justify your answer.

Question 2: Apply XPath for Data Selection - 10 Marks

Write suitable XPath expressions to extract each of the following: a.

All course records.

b. Names of all courses.

c. Courses having more than 50 students.

d. Courses carrying 4 credits.

e. Courses whose type is Theory.

f. Names of Theory courses having more than 50 students.

g. Faculty members handling courses with at least 4 credits.

- h. The course whose id is C104.
- i. The first course available in the XML document.
- j. The last course available in the XML document.

Question 3: Apply XSLT for Data Presentation - 10 Marks

Develop an XSLT stylesheet that transforms the XML data into an HTML table with the columns Course Code, Course Name, Faculty, Students, Credits, and Type. **a.** Display only courses having more than 40 students.

- b.** Sort the displayed courses in descending order of student enrollment.
- c.** Display a suitable heading such as “High Enrollment Courses”.
- d.** Use an XPath condition inside the XSLT transformation.
- e.** Produce valid HTML output.

Question 4: Interpret the Extracted Data - 5 Marks a.

Identify the course with the highest enrollment.

- b.** Identify the course with the lowest enrollment.
- c.** Determine the number of Theory courses.
- d.** Identify all courses having exactly 4 credits.
- e.** If an additional teaching assistant is assigned to every course with more than 60 students, identify the courses that require additional support.

Evaluation Rubric

Criterion	Marks
Correct interpretation of XML structure	5
Correct XPath expressions	10
Correct XSLT transformation and sorting	10
Correct interpretation of results	5
Total	30

Question 1: Interpret the XML Structure

a. Identify the root element.

The root element of the XML document is:

`<courses>`

It contains all the individual course records.

b. Identify the repeating record element.

The repeating record element is:

`<course>`

There are five course records: C101, C102, C103, C104 and C105.

c. Identify the attribute used to uniquely identify each course.

The attribute used to uniquely identify each course is:

`id`

Example:

`<course id="C101">`

d. Identify the elements that represent numeric information.

The numeric elements are:

`<students>`

`<credits>`

students represent the number of students enrolled, while credits represent the course credits.

e. State whether the XML document is structurally well-formed and justify your answer.

Yes, the XML document is well-formed.

Justification:

- It has a single root element `<courses>`.
- All elements have proper opening and closing tags.
- Elements are properly nested.
- The `id` attribute is properly enclosed in quotation marks.
- The XML declaration is correctly written.

The given dataset contains five <course> records under the <courses> root element.

Question 2: Apply XPath for Data Selection

```
<?xml version="1.0" encoding="UTF-8"?>
<courses>
  <course id="C101">
    <code>WEB301</code>
    <name>Web Technology</name>
    <faculty>Dr. Arun</faculty>
    <students>58</students>
    <credits>4</credits>
    <type>Theory</type>
  </course>
  <course id="C102">
    <code>AI302</code>
    <name>Artificial Intelligence</name>
    <faculty>Dr. Meena</faculty>
    <students>72</students>
    <credits>4</credits>
    <type>Theory</type>
  </course>
  <course id="C103">
    <code>WEB303</code>
    <name>Web Technology Laboratory</name>
    <faculty>Dr. Ravi</faculty>
    <students>36</students>
    <credits>2</credits>
    <type>Practical</type>
  </course>
  <course id="C104">
    <code>ML304</code>
    <name>Machine Learning</name>
    <faculty>Dr. Priya</faculty>
```

```
<students>64</students>
<credits>4</credits>
<type>Theory</type>
</course>
<course id="C105">
  <code>DB305</code>
  <name>Database Systems</name>
  <faculty>Dr. Kumar</faculty>
  <students>42</students>
  <credits>3</credits>
  <type>Theory</type>
</course>
</courses>
```

a. All course records

/courses/course

b. Names of all courses

/courses/course/name

c. Courses having more than 50 students

/courses/course[students > 50]

d. Courses carrying 4 credits

/courses/course[credits = 4]

e. Courses whose type is Theory

/courses/course[type = 'Theory']

f. Names of Theory courses having more than 50 students

/courses/course[type = 'Theory' and students > 50]/name

g. Faculty members handling courses with at least 4 credits

/courses/course[credits >= 4]/faculty

h. The course whose id is C104

/courses/course[@id = 'C104']

i. The first course available in the XML document

/courses/course[1]

j. The last course available in the XML document

/courses/course[last()]

These XPath requirements correspond to the selections specified in Question 2 of the paper.

Question 3: Apply XSLT for Data Presentation

The following XSLT transforms the XML data into an HTML table. It displays only courses with **more than 40 students** and sorts them in **descending order of student enrolment**, as required.

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet version="1.0"
  xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
  <xsl:output method="html" encoding="UTF-8"/>
  <xsl:template match="/">
    <html>
      <head>
        <title>High Enrollment Courses</title>
      </head>
      <body>
        <h1>High Enrollment Courses</h1>
        <table border="1">
          <tr>
            <th>Course Code</th>
            <th>Course Name</th>
            <th>Faculty</th>
            <th>Students</th>
            <th>Credits</th>
```

```

        <th>Type</th>
    </tr>
    <xsl:for-each select="courses/course[students > 40]">
        <xsl:sort select="students"
            data-type="number"
            order="descending"/>
        <tr>
            <td>
                <xsl:value-of select="code"/>
            </td>
            <td>
                <xsl:value-of select="name"/>
            </td>
            <td>
                <xsl:value-of select="faculty"/>
            </td>
            <td>
                <xsl:value-of select="students"/>
            </td>
            <td>
                <xsl:value-of select="credits"/>
            </td>
            <td>
                <xsl:value-of select="type"/>
            </td>
        </tr>
    </xsl:for-each>
</table>
</body>
</html>
</xsl:template>
</xsl:stylesheet>

```

(a) Display only courses having more than 40 students:

```
<xsl:for-each select="courses/course[students > 40]">
```


The important XPath condition is:

`courses/course[students > 40]`

So it excludes **WEB303**, because it has 36 students.

(b) Sort in descending order of student enrollment:

```
<xsl:sort select="students"
          data-type="number"
          order="descending"/>
```

So the output order becomes:

72

64

58

42

(c) Display a suitable heading

```
<h1>High Enrollment Courses</h1>
```

Output:

High Enrollment Courses

(d) Use an XPath condition inside XSLT

This is the XPath condition:

`courses/course[students > 40]`

It is used here:

```
<xsl:for-each select="courses/course[students > 40]">
```

So **(a)** and **(d)** are actually connected.

(e) Produce valid HTML output

The XSLT generates:

```
<html>
```

```
<head>
```

```
<title>High Enrollment Courses</title>
```

```
</head>
```

```
<body>
```

<h1>High Enrollment Courses</h1>

<table border="1">

and closes everything properly:

</table>

</body>

</html>

Also:

<xsl:output method="html"/>

tells the XSLT processor to produce HTML output.

Course Code	Course Name	Faculty	Students	Credits	Type
AI302	Artificial Intelligence	Dr. Meena	72	4	Theory
ML304	Machine Learning	Dr. Priya	64	4	Theory
WEB301	Web Technology	Dr. Arun	58	4	Theory
DB305	Database Systems	Dr. Kumar	42	3	Theory

Question 4: Interpret the Extracted Data

The given enrollment values are 58, 72, 36, 64 and 42.

a. Identify the course with the highest enrollment.

Artificial Intelligence (AI302) has the highest enrollment.

Number of students = 72

b. Identify the course with the lowest enrollment.

Web Technology Laboratory (WEB303) has the lowest enrollment.

Number of students = 36

c. Determine the number of Theory courses.

There are 4 Theory courses:

1. Web Technology
2. Artificial Intelligence
3. Machine Learning
4. Database Systems

d. Identify all courses having exactly 4 credits.

The courses with exactly 4 credits are:

1. Web Technology (WEB301)
2. Artificial Intelligence (AI302)
3. Machine Learning (ML304)

e. If an additional teaching assistant is assigned to every course with more than 60 students, identify the courses that require additional support.

Courses having more than 60 students are:

1. Artificial Intelligence (AI302) – 72 students
2. Machine Learning (ML304) – 64 students

Therefore, AI302 and ML304 require additional teaching assistant support.